Exam 4: Study Guide

This study guide lists skills you need to be able to do for your exam. It does not tell you *how* to do the skills. Skills may be tested individually, together, or not at all. It is up to you to determine what skills you need already know, and which skills you need more practice on. Review relevant topics in ALEKS, problems from the end of each chapter of your textbook, worksheets, and lab experiments.

Chapter 11 - Stoichiometry

1) Apply previously learned conversion factors as tools to get to moles, for use in stoichiometric calculations:
   a) Section 11.1
   b) Avogadro's number: counts of particles to moles
   c) Section 11.1.1 (Molar mass: mass to moles)
   d) Section 11.1.2 (Ideal Gas Law: Pressure, Volume, and Temperature of gas to moles)
   e) Section 11.1.2 (Molarity: volume of solution to mol)
2) Use the balanced chemical reaction as conversion factor to:
   a) Calculate amount of reactants needed to completely react with given amount of another reactant. (reactant to another reactant)
   b) Calculate amount of product that can form given the mass of one reactant (and excess of all other reactants). (reactant to product)
   c) Calculate amount of a reactant used given amount of product made. (product to reactant)
   d) Calculate amount of a product formed given amount of another product. (product to product)
   e) Note: "Amount" may need to be reported in mol, gram, count of particles, or volume (of gas or solution).
   f) $\boldsymbol{Some\ unit\ of\ A \rightarrow mol\ A \rightarrow mol\ B \rightarrow Some\ unit\ of\ B}$
3) Recognize that at constant pressure & temperature, volume of reactants/products will be proportional to stoichiometric coefficients.
   a) At STP, one mole of any gas will occupy 22.4 L
4) Determine the limiting reactant and excess reactant from sub-microscopic drawings of chemical reaction.
   a) Section 11.2 (one figure)
5) Given information about of two or more reactants, be able to calculate the theoretical yield of a chemical reaction.
   a) Section 11.2
   b) Determine limiting reactant mathematically.
6) Given actual yield, and theoretical yield, calculate the percent yield of a reaction.
   a) Section 11.3
7) Recognize the sign of enthalpy of reaction indicates whether a reaction is endothermic or exothermic.
   a) Section 11.4
8) Use the enthalpy of reaction (heat of reaction) as a conversion factor in calculations to determine amount of heat released or absorbed by a chemical reaction, or amount of reactants used to generate a specified amount of heat.
   a) Section 11.4

Chapter 12 – Acids and Bases

1) Describe properties of acids and bases
   a) Section 12.1.1, 12.1.2
2) Define Arrhenius acid/base, and Bronsted-Lowry acid/base
   a) Section 12.2.1
3) Based on chemical structure, identify compounds as most likely behaving as a base or an acid in solution. (Identify acid, base, conjugate acid, conjugate base in a reaction)
   a) Section 12.2.1
4) Predict the products of an acid reacting with a base
   a) Section 12.2.2
   b) Memorize which acids and bases are strong acids and bases
      i) Section 12.2.3
   c) Anything that isn't strong is weak
5) Describe acid/base strength in terms of degree of ionization
   a) Section 12.2.3
6) Use titration data to quantify amounts of acid or base in a reaction (stoichiometry)
   a) Section 11.1.3
7) Utilize the ionization product constant of water (Kw) to calculate the H+ or OH- concentration
   a) Section 12.3.1
      i) [Brackets] around a compound/ion mean the molarity of that compound/ion
8) Convert between [H+], [OH-], pH and pOH
   a) Section 12.3.2
      i) $[H^+][OH^-] = K_w = 1x10^{-14}$
      ii) $pH + pOH = 14$
      iii) $pH = -\log[H^+]$
9) Identify solutions as being acidic, basic, or neutral based on pH, pOH, [H+] or [OH-]
   a) Section 12.3.2
10) Identify which combinations of solutes would result in buffer solutions
   a) Section 12.4
11) Describe how a buffer solution resists change in pH
   a) Section 12.4

Chapter 13 – Chemical Equilibrium

1. Describe how concentration and temperature affects the rate of reaction based on collision theory.
2. Describe dynamic equilibrium in terms of rate of reaction and relative amounts of reactants and products.
    a. Section 13.1
3. Write the equilibrium constant expression (K_{eq}) for a balanced chemical equation
    a. Section 13.3
    b. **Remember:** pure liquids (l) and solids(s) do not appear in the equilibrium constant expression. Only aqueous (aq) and gaseous (g) compounds appear in the expression.
4. Interpret the magnitude of the equilibrium constant expression to determine if more products or reactants are favored
    a. Rank acids by relative strength given K_a.
        i. Section 13.6
5. Use molarities of reactants and products to calculate the equilibrium constant (K_{eq}) or use equilibrium constant expression to calculate molarity.
    a. Section 13.5
6. Apply Le Châtlier's principle to describe the response of a chemical system to a disturbance
    a. Section 13.4
    b. Addition or removal of reactants
        i. i.e. Removing reactant decreases number of collision between reactants → forward rate of reaction decreases. Rate of backwards reaction exceeds forward direction so more reactants form until equilibrium is re-established.
    c. Addition or removal of products
    d. Addition or removal volume
    e. Increase or decrease in temperature.
7. Describe the purpose of a catalyst and properties of a catalyst.

Chapter 16 – Nuclear Chemistry

1. Rank forms of nuclear radiation by ionizing power and penetration power
    a. Section 14.2
2. Write atomic symbols for elements by determining the number of neutrons and protons present
    a. Review Chapter 2
3. Write Balanced nuclear reactions for alpha decay, beta decay, gamma radiation, and positron emission
    a. Section 14.2
4. Determine how much of a sample is left given the half-life of a nuclide and time elapsed.
    a. The number of half-lives will always be in whole-number increments.
    b. Section 14.3
    c. Section 14.4.2
5. Define nuclear fusion and nuclear fission and write nuclear equations for each process
    a. Section 14.4.3